

SMI CRL microfocusing: initial feasibility

PROVISIONAL OPTICS MODEL - September 2026

Lens plane to sample at $Z=0$: 1350 mm
Z travel: -250 to 250 mm; positive downstream
Accessible image distance: 1100 to 1600 mm
Incident curvature at $Z=0$: 0 /m
Eight populated holders; 255 nonempty combinations evaluated.
Priority: fewest holders, then elements, then holder numbers.

Predicted gaps within 2.1-24 keV (analytic boundaries):
2.7619 - 3.4622 keV

ASSUMPTIONS

All currently labeled CRL holders are modeled as Be.
The 12 diamond lenses are unmodeled until their locations/radii are known.
Be density 1.848 g/cm³; diamond density 3.515 g/cm³.
Two parabolic surfaces per lens; quoted radii apply to each surface.
All holders share one plane.
Free-electron δ (Be $f_1=4$, C $f_1=6$); δ scales as energy⁻².
At 10 keV: $\delta(\text{Be}) = 3.405395\text{e-}06$; $\delta(\text{diamond}) = 7.29009\text{e-}06$.
Power: $P = \sum [2 \delta(\text{material}) N/R]$; focal length $f = 1/P$.
Ray curvature C: $1/q = 1/f - C(Z)$; $q = \text{sample_distance} - Z$.
 $C(Z) = C(0) / [1 + C(0) Z(\text{m})]$. Positive C means divergence.

INTERPRETATION

Coverage means a geometrical focus exists within nominal travel.
It does not establish spot size, transmission, or beam quality.
Material, radius convention, holder spacing and Z zero need confirmation. No tabulated dispersion correction is included.
Transmission needs lens aperture and web thickness; spot size also needs incident size/emittance and wavefront information.
Fewest holders can favor a thick stack over fewer lens elements.

Switches may require large Z jumps. Boundary solutions have zero travel margin; usable coverage will shrink with a margin.
A 1% focal-length error shifts Z by roughly 3.7-6.7 mm here.
Model and report generation issue no hardware commands.

Holder inventory and nominal positions

Holder	Contents	IN mm	OUT mm
-----	-----	-----	-----
1	1 x 50 um Be	0.0	10.0
2	8 x 50 um Be	0.0	10.0
3	16 x 50 um Be	0.0	10.0
4	4 x 50 um Be	0.0	10.0
5	2 x 50 um Be	0.0	10.0
6	empty	0.0	10.0
7	empty	0.0	-10.0
8	empty	0.0	-10.0
9	1 x 200 um Be	0.0	-10.0
10	8 x 500 um Be	0.0	-10.0
11	1 x 500 um Be	0.0	-10.0
12	2 mm aperture	0.0	-10.0

Empty holders 6-8 and aperture holder 12 have no lens power.

They are excluded from automatic lens selection.

IN=0 mm is nominal, pending alignment within about +/-2 mm.

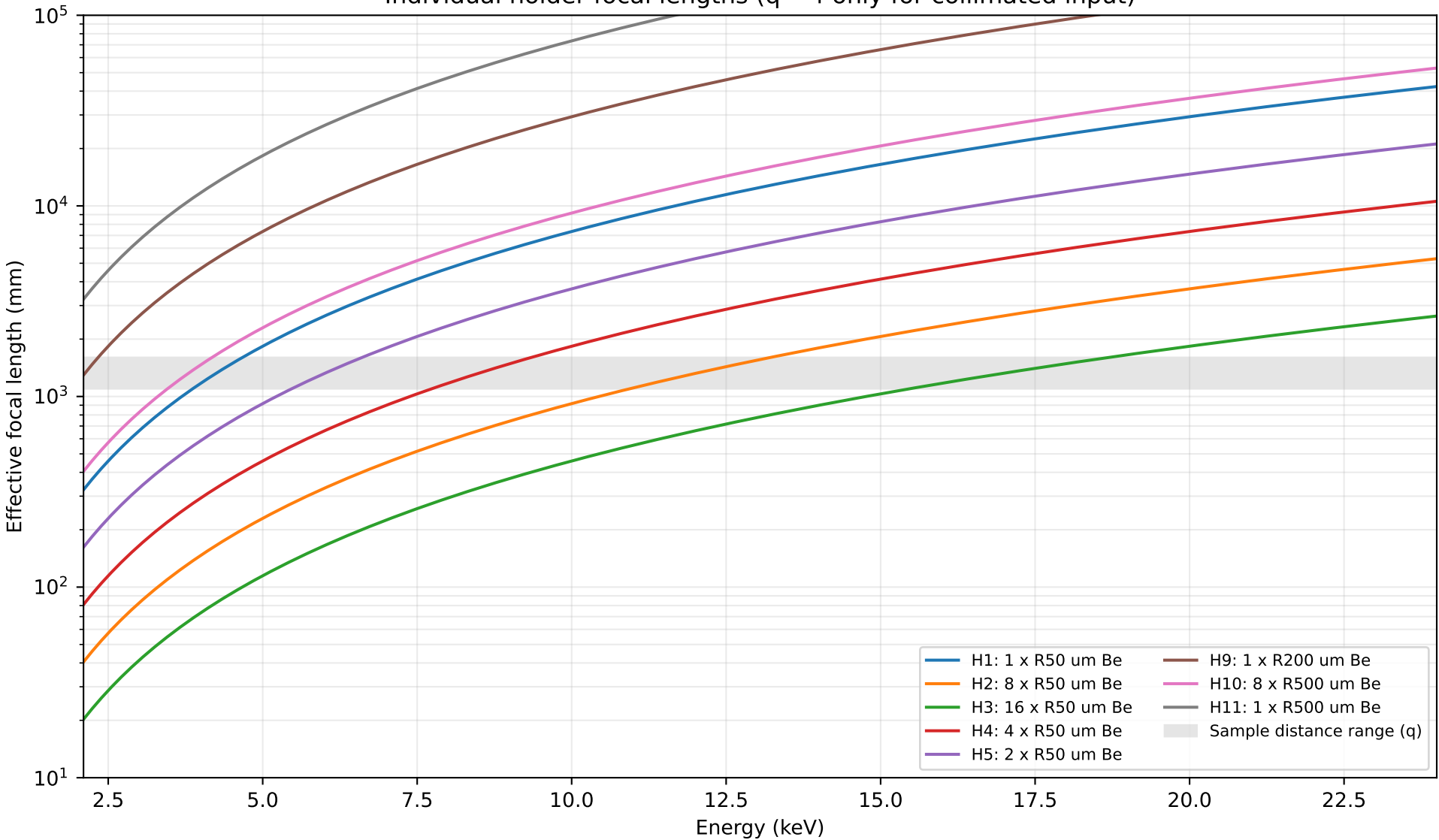
NEXT CALIBRATIONS / DECISIONS

1. Confirm material, per-surface radii, and holder axial spacing.
2. Measure at least one energy + holders + best-focus Z point.
3. Confirm actual motor travel and choose working end margins.
4. Choose alignment signal: transmission or measured spot size.
5. Decide aperture-12 behavior and persistence of IN positions.
6. Check horizontal/vertical curvature separately if astigmatic.

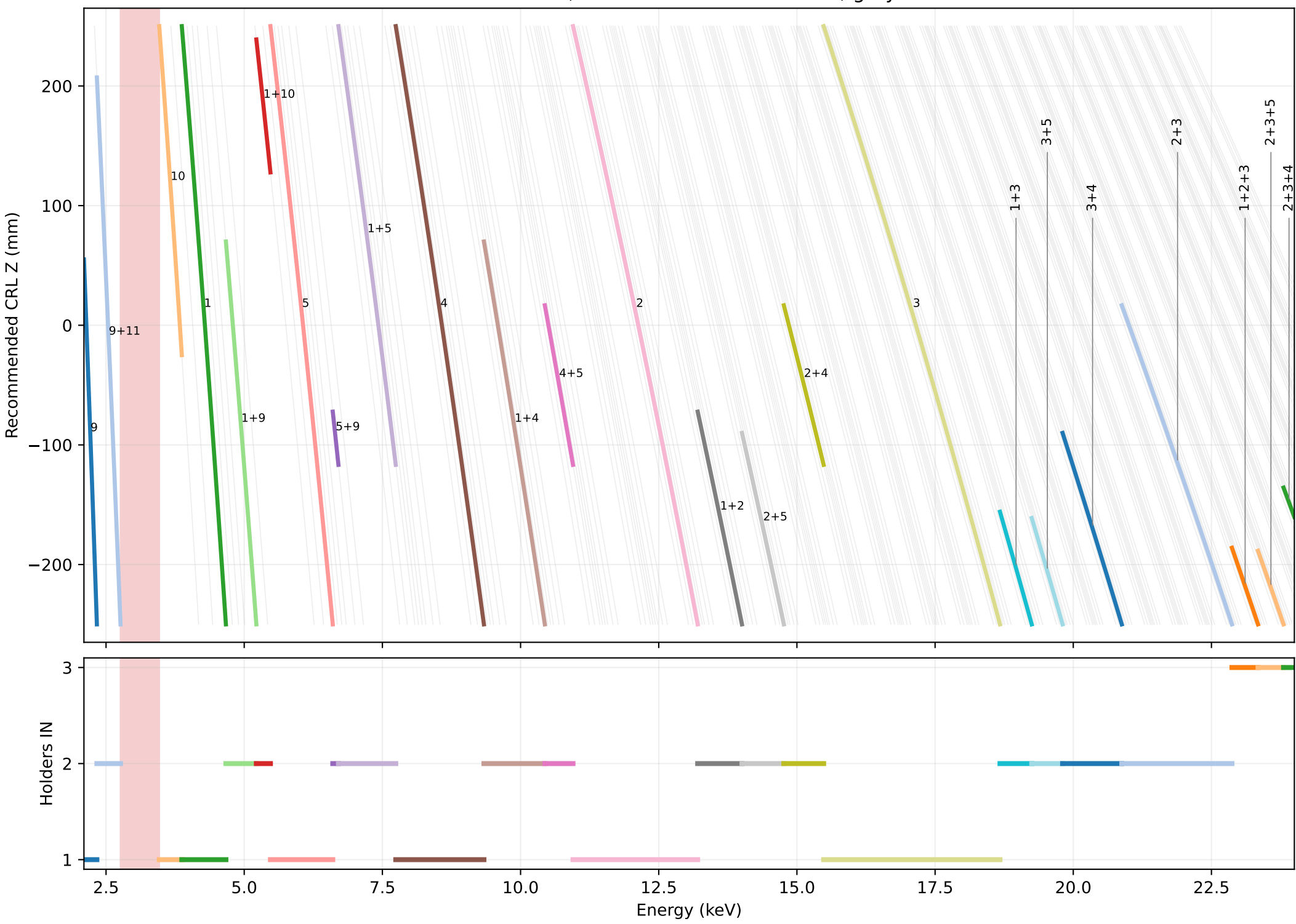
The existing lens1...lens12, x/y/z/ph/th remain motor components.

Initial CRL helpers expose inventory and offline calculations.

Individual holder focal lengths ($q = f$ only for collimated input)



Fewest-holder selection; labels are holder numbers; gray = alternatives



Recommended energy bands

Energy (keV)	Holders IN	N	Z start -> Z end (mm)
2.100- 2.334	9	1	55.0 -> -250.0
2.334- 2.762	9+11	2	207.1 -> -250.0
2.762- 3.462	UNREACHABLE		
3.462- 3.871	10	8	250.0 -> -25.0
3.871- 4.668	1	1	250.0 -> -250.0
4.668- 5.219	1+9	2	70.0 -> -250.0
5.219- 5.474	1+10	9	238.9 -> 127.8
5.474- 6.602	5	2	250.0 -> -250.0
6.602- 6.705	5+9	3	-72.2 -> -116.7
6.705- 7.742	1+5	3	250.0 -> -116.7
7.742- 9.337	4	4	250.0 -> -250.0
9.337- 10.439	1+4	5	70.0 -> -250.0
10.439- 10.949	4+5	6	16.7 -> -116.7
10.949- 13.204	2	8	250.0 -> -250.0
13.204- 14.005	1+2	9	-72.2 -> -250.0
14.005- 14.763	2+5	10	-90.0 -> -250.0
14.763- 15.484	2+4	12	16.7 -> -116.7
15.484- 18.674	3	16	250.0 -> -250.0
18.674- 19.249	1+3	17	-155.9 -> -250.0
19.249- 19.807	3+5	18	-161.1 -> -250.0
19.807- 20.878	3+4	20	-90.0 -> -250.0
20.878- 22.871	2+3	24	16.7 -> -250.0
22.871- 23.342	1+2+3	25	-186.0 -> -250.0
23.342- 23.805	2+3+5	26	-188.5 -> -250.0
23.805- 24.000	2+3+4	28	-135.7 -> -160.2

Endpoints rounded; use selection_bands.csv for precision.

At shared endpoints both adjacent choices can be reachable.

Use recommend() at the exact energy, not interpolation across a holder change. N is the total number of individual lenses.

Sensitivity to incident curvature, holding baseline lens choice fixed

